GLINT : PERSONALIZED GIFT COLLECTION

This project involves the development of a single-product e-commerce platform focused on personalized gifting. Unlike conventional e-commerce sites that offer fixed, pre-assembled products, this platform enables users to construct a customized gift box tailored to the recipient and the occasion.Users may add or remove items from a selected gift box, apply personalized text such as names, quotes, or messages, and view real-time price updates based on their choices. The platform also includes cart management, secure checkout, and payment processing.

1. INTRODUCTION

1.1. PURPOSE

The purpose of this project is to develop an interactive e-commerce platform that enables users to create fully personalized gift boxes. The system allows users to select a gift box based on an occasion, customize its contents by adding or removing items, personalize it with names or messages, and complete the purchase through a secure checkout process. The platform aims to provide a transparent, user-centric gifting experience where pricing adjusts dynamically based on user choices.

1.2. DOCUMENT CONVENTION

This document is structured according to standard Software Requirements Specification (SRS) formatting, following IEEE guidelines. Within this specification, the term "Must" or "Shall" denotes a mandatory requirement that must be implemented, while "Should" indicates a recommended but non-mandatory feature. Key terms are defined as follows: a "Gift Box" refers to a curated collection of physical items presented in a single package; an "Event" is a predefined occasion that determines which gift boxes are displayed to users; "Customization" describes the act of adding, removing, or personalizing items within a gift box; and "Base Price" is defined as the standard cost of a gift box with its default set of items.

1.3. INTENDED AUDIENCE AND READING SUGGESTIONS

This document is intended for four primary audiences: project developers, who will rely on it to understand the functional and technical requirements; project

managers, who will use it for project planning and scope management; the quality assurance team, for test case design and validation; and stakeholders, to review overall project objectives and deliverables. To help each group navigate the material efficiently, the following reading suggestions are provided: technical team members should focus on sections 3 through 5, project managers are advised to review sections 1 and 2, and stakeholders are recommended to read the introduction and the overall description sections for a high-level overview.

1.4. PROJECT SCOPE

The scope of this project encompasses user registration, authentication, and profile management; the display of gift boxes filtered by user-selected events; real-time customization of gift boxes including adding or removing items and adding personalized text with dynamic price recalculation based on those customizations; shopping cart management and order checkout; and payment gateway integration for transaction processing. However, this project is limited to a web-based platform and explicitly does not include a mobile application or a physical inventory management system.

1.5. REFERENCES

The following documents and resources are referenced in this SRS:

- Python Django Official Documentation (v4.2+)
- Bootstrap Framework Documentation
- Payment Gateway (e.g., Stripe, PayPal) API Documentation
- OWASP Security Guidelines for Web Applications
- W3C Web Content Accessibility Guidelines (WCAG) 2.1

2. OVERALL DESCRIPTION

2.1. PRODUCT PERSPECTIVE

This system is a standalone web application built using the Python Django framework, operating as a new, self-contained product that does not replace or interface with any existing systems. The application interacts with several external components: a PostgreSQL or MySQL database for persistent storage of user data, product inventory, order history, and customization details; the file system to store user-uploaded images or text files such as custom letters as part of the gift box customization; an external payment gateway to process financial transactions securely; and an email service to send order confirmations and

notifications. The system is accessed via standard web browsers and is hosted on a cloud platform.

2.2. PRODUCT FEATURES

The key features of the platform are:

User Authentication
User authentication encompasses secure registration and login functionality, along with password encryption and recovery mechanisms to ensure account security and user access management.

Event-Based Browsing
Event-based browsing allows users to browse gift recommendations by selecting a specific occasion such as Birthday, Anniversary, or Valentine's Day and in response, the system presents a curated selection of items that are specifically suggested for that chosen event.

Gift Box Customization
Gift box customization is facilitated through an interactive, step-by-step builder interface that guides users through the personalization process. Within this interface, users can add items to their gift box, remove unwanted ones, and further personalize the box by including the recipient's name, a custom quote, or a personal letter entered as free text. To support informed decision-making, the system incorporates real-time pricing, dynamically calculating and displaying the updated total price as users add or remove items, ensuring full transparency throughout the customization experience.

Shopping Cart
The shopping cart functionality allows users to add customized gift boxes to their cart, with the ability to view, edit, or remove any items before proceeding to checkout. Additionally, the system ensures cart persistence across sessions, so that users can return to their shopping cart at a later time without losing their selected items and customizations.

Payment Processing
Payment processing includes integration with a secure payment gateway to handle financial transactions, supporting multiple payment methods such as credit and debit cards, UPI, and other commonly used options. Upon successful

completion, the system provides transaction confirmation and generates a receipt for the user's records.

Order Management
Order management enables users to view their complete order history and track the current status of each order. On the administrative side, the system allows admins to manage orders by performing actions such as marking items as shipped, updating statuses, and handling fulfillment workflows, ensuring efficient oversight of the order lifecycle.

2.3. USER CLASSES AND CHARACTERISTICS

The system defines three primary user classes based on role and authentication status.
A Guest User is an unauthenticated visitor who can browse items, view event pages, and use the customization builder, but cannot finalize orders.
A Registered Customer is an authenticated user with an account, who enjoys full purchasing capabilities including saving gift boxes, adding items to the cart, completing purchases, viewing order history, and managing their profile.
An Admin is an administrative user responsible for platform management, with privileges to manage inventory, view and update orders, handle user accounts, and oversee system operations.

2.4. OPERATING ENVIRONMENT

The platform is a web-based application that runs on the server side and is compatible with Windows, macOS, and Linux operating systems. For production deployment, it uses Gunicorn or Nginx as the web server, while the Django development server is used during development. The database is PostgreSQL in production and SQLite in development. The application supports all modern browsers, including Chrome, Firefox, Safari, and Edge, and requires internet connectivity for full functionality.

2.5. DESIGN AND IMPLEMENTATION CONSTRAINTS

The design and implementation of the system are governed by several specific constraints. The technology stack shall be built using Python Django (version 4.2 or higher) alongside HTML5, CSS3, and JavaScript, with Bootstrap 5 used as the front-end framework to ensure responsive design. For production, PostgreSQL shall be the designated database, and the system shall integrate with a

third-party payment gateway such as Stripe or PayPal. Security measures mandate that all sensitive data including passwords and payment information be encrypted, and HTTPS shall be enforced throughout. Code quality is maintained by following PEP 8 guidelines for all Python code. Additionally, the system must be fully functional across mobile, tablet, and desktop screens, and user sessions shall be managed server-side with secure cookie handling.

2.6. USER DOCUMENTATION

The project will include comprehensive user documentation to support all audiences. A User Manual will provide a step-by-step guide for using the platform, covering everything from registration to checkout. An Admin Manual will detail administrative procedures and system management tasks for platform oversight. Developer Documentation will offer technical documentation to facilitate maintenance, troubleshooting, and future updates. Additionally, the system will feature Online Help in the form of in-app contextual help and tooltips to guide users through specific tasks, alongside an FAQ Section on the website that addresses frequently asked questions for quick reference.

2.7. ASSUMPTIONS AND DEPENDENCIES

The project is built on a set of key assumptions and dependencies that must be in place for successful operation. It assumes that users have reliable internet connectivity, possess basic computer literacy, and hold valid payment methods, while also relying on the availability and proper functioning of payment gateway services, as well as maintained database and server infrastructure. On the dependency side, the system depends on the Django framework and Python runtime environment being available, the database system being operational, the payment gateway API being accessible, email services being reliable for sending notifications, and CDN services being available for serving static files efficiently.

3. SYSTEM FEATURES

USER REGISTRATION

The User Registration Module serves as the primary entry point for new users to create an account on the Personalized GiftBox Collection Platform, enabling guest users to sign up by providing their Full Name, a valid Email address, and a secure Password with confirmation. The system performs comprehensive input validation, ensuring the email format is correct and the password meets strength

requirements of at least 8 characters containing both letters and numbers. Upon submission, the module verifies that the email is not already registered in the database and that both password entries match exactly. All passwords are securely hashed using Django's built-in hashing algorithms such as Argon2 or PBKDF2 before storage, ensuring plaintext passwords are never saved. After successful validation and account creation, the system automatically dispatches a confirmation email to the user's registered address containing a verification link for account activation. The module includes robust error handling mechanisms that display user-friendly messages for common issues like duplicate email addresses, weak passwords, or mismatched entries, guiding users to correct their input without exposing technical system details. The registration flow seamlessly integrates with the login system, allowing new users to proceed directly to authentication once their account is successfully created, and also handles edge cases such as email service unavailability by logging errors and notifying users appropriately while still allowing account creation.

USER LOGIN

The User Login Module provides secure authentication for registered users to access their accounts and utilize the full functionality of the Personalized GiftBox Collection Platform. The module presents a login form requiring users to enter their registered Email address and Password, which are then validated against the stored credentials in the database. The system incorporates robust security measures including password hashing verification, session management with secure cookie handling using HttpOnly and Secure flags, and automatic session timeout after 30 minutes of inactivity to protect user accounts from unauthorized access. To prevent brute force attacks, the module implements an account lockout mechanism that temporarily restricts access after five consecutive failed login attempts, displaying an appropriate error message and directing users to the password reset functionality. Upon successful authentication, the system creates a user session, redirects users to their dashboard or the homepage, and displays personalized content based on their profile. The module also supports a "Remember Me" feature that creates a persistent session cookie for users who wish to stay logged in across browser sessions. For users who have forgotten their credentials, the module provides a seamless integration with the password reset functionality through a "Forgot Password" link. Additionally, the login system includes comprehensive error handling that displays user-friendly messages for invalid credentials, locked accounts, or database connectivity issues, ensuring users understand the reason for authentication failures without exposing

sensitive system information, while also maintaining detailed logs for monitoring and security auditing purposes.

GIFTBOX CUSTOMIZATION

The Gift Box Customization Module serves as the core feature of the Personalized GiftBox Collection Platform, providing users with an interactive step-by-step builder interface to create fully personalized gift boxes tailored to their recipients and occasions. The module guides users through a structured three-step process beginning with item selection, where users can browse available products and add items to their gift box with the ability to specify multiple quantities for each item, and also remove items or adjust quantities as needed while viewing a visual preview of their creation. The system dynamically updates the item list in real-time and calculates the running total as users make changes to their selections. The second step focuses on personalization, requiring users to enter the recipient's name as a mandatory field while providing optional fields for adding custom quotes and personal letters, with all text inputs undergoing validation to ensure appropriate content. The third step presents a comprehensive review screen displaying a complete summary of the gift box including all selected items with quantities and prices, personalization details, and the total cost breakdown. Throughout this process, the module maintains real-time price calculation that instantly updates the total price whenever items are added, removed, or quantities are modified, ensuring complete pricing transparency for users. The module also includes a draft saving functionality that allows users to save incomplete gift boxes for later completion, with registered users having their drafts permanently stored in the database while guest users have drafts stored temporarily in their session. Users can proceed to add their finalized gift box to the shopping cart or continue editing, and the module includes comprehensive validation ensuring that gift boxes contain at least one item and have a recipient name before allowing progression, while also handling exceptions such as session expiry, item unavailability, and validation errors with appropriate user-friendly messages.

SHOPPING CART

The Shopping Cart Module enables users to aggregate and manage their customized gift boxes before proceeding to checkout, providing a centralized location for reviewing and modifying their selections. Once users finalize their gift box customization, they can add it to the cart with a single click, and the system immediately updates the cart count displayed in the header while storing all gift box details including selected items with quantities, personalization information,

pricing breakdown, and customization data. The cart page presents users with a comprehensive list of all gift boxes, each displaying its itemized contents, personalization details, individual price, and the ability to edit or remove specific gift boxes entirely. Users can modify their selections directly from the cart by clicking an edit button that opens the builder interface with the saved state, allowing them to add or remove items, update personalization, and adjust quantities, with all changes automatically reflected in the cart upon saving. The system performs real-time calculations on the cart page, displaying the subtotal for each gift box, applicable taxes, shipping costs if applicable, and the grand total, with all values updating dynamically as users make modifications. For authenticated users, cart contents persist across sessions through database storage, ensuring their selections remain available even after logging out and returning later, while guest users have their cart maintained via session storage, with the system seamlessly merging guest cart contents upon user registration or login. The module also handles edge cases such as displaying appropriate messages when the cart is empty, showing a disabled checkout button, guiding users to continue shopping, and managing situations where items become unavailable by notifying users and automatically removing affected items from the cart. Additionally, the cart module integrates with the checkout system, providing users with a clear path to proceed to payment while maintaining accurate inventory tracking and preventing duplicate additions through prompts asking users whether they want to add another copy or update existing cart items.

PAYMENT AND CHECKOUT

The Payment and Checkout Module enables authenticated users to securely complete their purchases by processing financial transactions through integrated third-party payment gateways such as Stripe or PayPal. When users click the "Proceed to Checkout" button from their shopping cart, the system first verifies authentication status and redirects guest users to login or registration pages before allowing them to continue. The checkout page displays a comprehensive order summary showing all gift boxes with their items, personalization details, quantities, individual prices, and the complete cost breakdown including subtotal, applicable taxes, and grand total, along with pre-filled shipping address information retrieved from the user's profile for convenience. Users can review their entire order, select their preferred payment method, and confirm their purchase with a single click, at which point the system redirects them to the secure payment gateway interface where they can safely enter their payment credentials without any sensitive financial data being stored on the local server, ensuring full PCI DSS compliance. The module handles payment success

callbacks by automatically marking orders as "Paid", generating unique order confirmation numbers, sending detailed confirmation emails to users containing their order summary and tracking information, clearing the user's cart, and displaying a success page with the order details. In cases of payment failure, the system displays appropriate error messages explaining the reason for decline and allows users to retry using alternative payment methods or the same method after correcting any issues. The module implements atomic transaction processing to ensure data integrity by rolling back all changes if any step in the order creation process fails, preventing partial order creation or inconsistent data states. Additionally, the system handles various edge cases including payment gateway timeouts with appropriate user notifications, cart emptiness during checkout attempts, and service unavailability scenarios with graceful error handling and logged exceptions for monitoring purposes, while also supporting multiple currencies based on deployment region requirements.

EVENT-BASED BROWSING

The Event-Based Browsing Module allows users to explore gift box suggestions organized by occasion or event type, providing context-aware recommendations that streamline the gift selection process. The module displays a list of predefined events such as Birthday, Anniversary, Valentine's Day, Mother's Day, and other special occasions on the homepage with engaging visual cards that capture the theme of each event. When users click on a specific event, the system fetches and displays curated gift boxes associated with that occasion, presenting products in a grid format with images, prices, brief descriptions, and an "Add to Builder" button on each product card to enable quick customization. The module includes filtering capabilities that allow users to refine their browsing experience by product categories such as Chocolates, Flowers, Accessories, and other relevant classifications, with the system updating displayed products instantly as filters are applied or removed. Users can click on individual products to view detailed information including comprehensive descriptions, image galleries, availability status, and pricing, which helps them make informed decisions about which items to include in their customized gift boxes. The event-based recommendations are fully configurable by administrators, allowing them to associate products with multiple events, create new event categories, remove existing ones, and curate which products appear under each occasion to ensure relevance and seasonal appropriateness. The module also handles scenarios where no products are available for a particular event by displaying appropriate user-friendly messages and suggesting alternative occasions, while maintaining fast response times through optimized database queries with proper indexing for sub-100ms read operations even with extensive product catalogs.

Additionally, the browsing interface is designed responsively to work seamlessly across desktop, tablet, and mobile devices, ensuring a consistent experience regardless of the device used to access the platform, and integrates smoothly with the gift box builder so users can seamlessly transition from browsing to customization.

ADMIN-PRODUCT MANAGEMENT

The Admin Product Management Module provides administrators with comprehensive control over the platform's product inventory, enabling them to add, edit, and delete product items as well as manage all associated product data. Through the admin dashboard, administrators can create new products by filling in detailed information including product name, description, price, stock quantity, and assigning products to multiple categories, while uploading product images with validation ensuring images are optimized for web display and under 5MB in size. The module allows administrators to edit existing products by updating any field including changing prices, modifying descriptions, updating stock quantities, adding or removing images from the product gallery, and reassigning category associations, with all changes validated and saved to the database upon confirmation. Products can be deleted from the system with a confirmation step to prevent accidental removal, and when products are deleted, the system handles dependencies by checking for existing associations with orders or active cart items before proceeding. The product management interface includes search and filter functionality that enables administrators to quickly locate specific products by name, category, price range, or stock status, and supports sorting options for efficient inventory management across potentially large product catalogs. The module also implements validation rules ensuring product prices are positive numbers, product names are unique to prevent duplicates, and all required fields are completed before allowing product creation, with user-friendly error messages displayed for any validation failures. Additionally, administrators can manage product visibility by enabling or disabling products without deleting them, useful for seasonal items or temporary unavailability, and the system maintains audit logs of all product management actions for accountability and tracking purposes, while the module integrates seamlessly with the event-based browsing system to ensure products are properly categorized and displayed to users based on selected occasions.

ADMIN-ORDER MANAGEMENT

The Admin Order Management Module enables administrators to view, process, and manage all customer orders placed on the Personalized GiftBox Collection

Platform through a comprehensive dashboard interface. The module presents a complete list of all orders displaying essential information including order ID, customer name, order date, total amount, and current status, with filtering capabilities that allow administrators to sort orders by status categories such as Pending, Paid, Shipped, and Completed, as well as filter by date ranges to view orders from specific periods. Administrators can click on any order to access detailed information including the complete order breakdown with itemized product lists showing quantities and prices, customer personalization details such as recipient names and custom messages, shipping addresses, payment confirmation details, and order history logs. The module provides functionality to update order statuses through a workflow that guides orders from Pending to Paid to Shipped to Completed, with each status change triggering automated email notifications to customers informing them of their order progress and including relevant tracking information when applicable. When orders need to be cancelled, administrators can initiate the cancellation process with confirmation steps, and the system automatically handles refund processing when payments have already been completed. The module includes additional features such as adding internal notes to orders for team communication, printing order invoices and packing slips for fulfillment, and managing bulk status updates for processing multiple orders efficiently during peak periods. Comprehensive search functionality enables administrators to quickly locate specific orders by order ID, customer email, or customer name, and the module integrates with inventory management by automatically updating stock quantities when orders are confirmed. The interface also provides dashboard statistics showing order volumes by status, helping administrators identify bottlenecks and manage fulfillment workflows effectively, while maintaining detailed audit logs of all order management actions including who made changes and when, ensuring accountability and traceability for all order processing activities. Additionally, the module handles edge cases such as displaying appropriate messages when no orders match the applied filters, managing incomplete order data gracefully, and providing error handling for update failures with user-friendly messages and retry options.

ORDER HISTORY

The Order History Module enables registered customers to view and track all their past purchases made on the Personalized GiftBox Collection Platform through a dedicated dashboard section. When users navigate to their order history, the system retrieves and displays a comprehensive list of all orders associated with their account, showing key information for each order including the unique order number, date placed, total amount paid, and current status with

visual indicators such as Pending, Paid, Shipped, or Completed. Users can click on any order to access a detailed view that presents the complete order breakdown including all items with quantities and individual prices, personalization details such as recipient names and custom messages, shipping address information, payment confirmation details, and any tracking numbers or shipment updates provided by the admin. The module includes filtering and sorting capabilities that allow users to search for specific orders by date ranges or status, making it easy to locate particular purchases among multiple orders. For orders that are still in progress, users can monitor the status changes in real-time as administrators update the order through the fulfillment process, providing transparency and peace of mind about their purchase status. The module also includes functionality for users to reorder previous gift boxes by clicking a convenient reorder button that automatically loads the items and personalization details from their past order into the gift box builder, allowing them to quickly recreate similar gifts for other occasions or repeat purchases. Users can print order receipts and order summaries directly from the detailed view page, which is useful for expense tracking, gift records, or returns. Additionally, if the platform supports order cancellation policies, the module provides appropriate actions for cancelling eligible orders with clear confirmation dialogs and notifications about refund processing when applicable. The system also handles scenarios where users have no orders by displaying a friendly message encouraging them to start shopping, while maintaining fast response times through optimized database queries and implementing proper security measures to ensure users can only view and access their own order data, protecting sensitive order information from unauthorized access.

SAVED DRAFTS

The Saved Drafts Module enables users to save incomplete gift box customizations for later completion, providing flexibility and convenience in the gift creation process. When users are in the Gift Box Builder and need to pause their customization, they can click the "Save Draft" button, and the system stores the current state including all selected items with quantities, personalization details such as recipient name, quotes, and messages, and pricing information, creating a snapshot of their progress. Registered users have their drafts permanently stored in the database, allowing them to access their saved drafts from any device across multiple sessions, while guest users have drafts maintained temporarily in their session storage, and the system prompts them to register when attempting to save drafts permanently. Users can access their saved drafts through a dedicated section in their dashboard or profile area, where the system presents a list of all saved drafts displaying essential

information such as the draft name or creation date, the number of items in the box, and the last modified timestamp, with sorting options to organize drafts by date or name. From the drafts list, users can click on any draft to load it directly into the gift box builder, resuming customization exactly where they left off with all selections and personalization intact, and they can complete the gift box, save further changes, or delete drafts that are no longer needed. The module also allows users to rename drafts for better organization, making it easier to manage multiple drafts created for different recipients or occasions. The system includes a draft retention policy ensuring drafts are recoverable for at least 30 days, with automatic cleanup of abandoned drafts to maintain database efficiency, and handles edge cases such as session expiry warnings prompting users to save their work before proceeding. Additionally, when items within a saved draft become unavailable or out of stock, the system displays appropriate notifications to users upon loading the draft, informing them about the unavailability and allowing them to make alternative selections, ensuring the customization process remains smooth and transparent. The module integrates seamlessly with the authentication system, so when guest users who have temporary drafts finally register or log in, the system intelligently migrates their session-based drafts to their new permanent account, ensuring no progress is lost during the transition from guest to registered user, thereby enhancing the overall user experience and encouraging account creation without disrupting the customization workflow.

ADMIN ANALYTICS

The Admin Analytics Module provides administrators with comprehensive insights and performance metrics about the Personalized GiftBox Collection Platform through an interactive dashboard that displays real-time and historical data. The module presents key performance indicators including total orders placed, total revenue generated, user registration trends, and average order value, all displayed through visually appealing charts and graphs that make data interpretation intuitive and actionable. Administrators can filter analytics by custom date ranges such as daily, weekly, monthly, or yearly periods to analyze trends over time and identify seasonal patterns in purchasing behavior. The module includes product performance analytics showing the most popular gift box items, best-selling categories, and inventory turnover rates, helping administrators make informed decisions about product stocking and marketing strategies. Order analytics provide insights into order volume trends, peak ordering times, geographic distribution of customers, and conversion rates from browsing to purchase, enabling administrators to optimize the user experience and identify potential bottlenecks in the sales funnel. The dashboard also tracks user behavior metrics such as registration rates, cart abandonment rates, and

the popularity of different event categories, offering valuable feedback on platform usability and customer preferences. Administrators can export analytics data in various formats including CSV, PDF, or Excel for further analysis, reporting, or presentation to stakeholders, supporting data-driven decision-making and strategic planning. The module displays revenue breakdowns by different product categories and events, helping identify which segments generate the most income and which may require promotional efforts to boost sales. Additionally, the analytics system is designed for performance efficiency with data cached appropriately to ensure fast loading times even when processing large datasets, and the module maintains historical data for trend comparison, allowing administrators to benchmark current performance against past periods and track the impact of platform changes or marketing campaigns. The analytics dashboard also includes alerts and notifications for significant changes in metrics such as sudden drops in conversion rates or unusual order volumes, enabling administrators to quickly investigate and address potential issues that could impact business performance or user satisfaction.

4. EXTENDED INTERFACE REQUIREMENTS

4.1. USER INTERFACES

The user interface design centers on a responsive web interface that shall be compatible with all modern browsers and devices, including desktop, tablet, and mobile. The design language employs a clean, minimalistic aesthetic with intuitive navigation, using warm and inviting primary colors to reflect the gifting theme. The homepage shall display event categories with engaging visuals to guide browsing. The builder interface follows a step-by-step wizard featuring a progress indicator which includes Step 1: Select Items, Step 2: Personalize, Step 3: Review alongside a visual preview of the gift box and real-time price display. The cart page shall show a summarized list of items, including customization details and the total price, while the checkout page includes a payment button with a secure redirect. The user dashboard displays order history, profile settings, and saved drafts, and the admin dashboard uses a custom panel with full CRUD interfaces for managing products, categories, orders, and users.

4.2. HARDWARE INTERFACES

The system is designed to operate on standard server hardware, specifically cloud-based infrastructure with a minimum configuration of 2 vCPUs, 8 GB of

RAM, and 50 GB of storage. On the client side, the application shall be accessible via standard devices such as PCs, laptops, tablets, or smartphones, all equipped with a modern browser and an active internet connection. Importantly, no specialized hardware interfaces are required, ensuring broad accessibility and ease of deployment.

4.3. SOFTWARE INTERFACES

The system interfaces with several external software components, each serving a distinct purpose and using a specific protocol. The PostgreSQL database handles the storage and retrieval of all application data including users, products, orders, and customizations—via SQL through the Django ORM. The payment gateway (Stripe or PayPal) processes financial transactions through a REST API secured over HTTPS. The email service uses the SMTP protocol to send order confirmations, password reset emails, and other notifications. For file storage, the system stores user-uploaded images and text files either on the local filesystem or via AWS S3. Finally, the web browser renders the front-end interface and communicates with the application using standard HTTP or HTTPS protocols.

4.4. COMMUNICATION INTERFACES

All communication between the client and the server shall be conducted over HTTPS to ensure data security, with RESTful endpoints used for interacting with the payment gateway and other third-party services. Email communication shall rely on the SMTP protocol, and the application shall run on port 443 for HTTPS traffic, with port 80 optionally available for HTTP redirection as needed.

5. OTHER NONFUNCTIONAL REQUIREMENTS

5.1. PERFORMANCE REQUIREMENTS

To ensure a smooth and reliable user experience, the system establishes clear performance benchmarks: page load times shall remain under three seconds for 95% of all requests, while the platform must support at least 500 concurrent users without performance degradation. During peak traffic, the system shall process a minimum of 100 orders per minute, and it should be designed for horizontal scalability by adding more application servers as demand grows. Additionally, database performance is critical, with all read operations optimized through proper indexing to guarantee sub-100ms response times.

5.2. SAFETY REQUIREMENTS

To ensure reliability and data integrity, the system mandates automated database backups on a daily basis, with a comprehensive disaster recovery plan in place to restore full system functionality within four hours of any failure. All financial and data transactions shall be atomic, meaning that in the event of a failure, the system will automatically roll back to its previous consistent state to prevent partial or corrupted updates. Furthermore, error handling shall be user-friendly, displaying clear and helpful messages without exposing any underlying technical details or stack traces to the end user.

5.3. SECURITY REQUIREMENTS

The security requirements for the system are comprehensive and span multiple layers of protection. For authentication, passwords shall be hashed using strong algorithms such as Argon2 or PBKDF2. Session cookies shall be secured with the HttpOnly and Secure flags to prevent client-side script access and ensure transmission only over encrypted channels. In terms of payment security, the system shall comply with PCI DSS standards to safeguard all financial transactions. Input validation is mandatory across all user inputs, with strict sanitization to prevent SQL injection, cross-site scripting (XSS), and cross-site request forgery (CSRF) attacks. Role-based access control (RBAC) shall enforce distinct permissions between admin and regular users, restricting access to sensitive operations accordingly. Additionally, all sensitive user data—including names and addresses shall be encrypted at rest, and the system shall implement essential security headers such as Content Security Policy (CSP), HTTP Strict Transport Security (HSTS), and X-Frame-Options to mitigate common web vulnerabilities.

5.4. SOFTWARE QUALITY ATTRIBUTES

The system is designed with a strong emphasis on quality attributes across multiple dimensions. For reliability, it shall achieve 99.9% uptime, excluding any scheduled maintenance windows. Usability is prioritized with an intuitive interface that requires a minimal learning curve, targeting a System Usability Score (SUS) of 80 or higher. Maintainability is ensured by adhering to Django's best practices and PEP 8 guidelines, with comprehensive documentation embedded throughout the codebase. Portability allows deployment on any platform that supports Python and Django, including Linux, macOS, and Windows. Testability is

addressed through unit tests covering at least 80% of the backend code, alongside integration tests that cover all critical user flows. Finally, accessibility compliance is mandated by conforming to WCAG 2.1 AA standards to ensure the platform is usable by individuals with disabilities.

6. OTHER REQUIREMENTS

6.1. Legal and Regulatory Compliance

The system shall adhere to all relevant legal and regulatory requirements. For data privacy, it will comply with applicable laws such as the GDPR and CCPA, which includes obtaining explicit user consent for data collection and honoring users' rights to request deletion of their personal data. Payment compliance is addressed by meeting PCI DSS Level 3 requirements for secure handling of financial information. Additionally, a terms of service document shall be clearly presented to users during the registration process to ensure informed consent and legal transparency.

6.2. Development Environment

The development environment is structured to ensure efficiency, quality, and collaboration. Version control shall be managed using Git, with the repository hosted on a platform such as GitHub or GitLab. A continuous integration and deployment (CI/CD) pipeline shall be established using tools like GitHub Actions or Jenkins to automate testing and deployment processes. For testing, the system shall employ Django's built-in test framework alongside Pytest for both unit and integration tests. All project dependencies shall be explicitly listed in a `requirements.txt` file to facilitate consistent environment setup and dependency management across development and production stages.

6.3. Deployment Requirements

The deployment requirements specify that the system shall be hosted on a major cloud platform such as AWS, Azure, or Google Cloud to ensure scalability and reliability. All sensitive credentials, including API keys and database passwords, shall be stored securely as environment variables rather than hard-coded in the source code. Additionally, the system shall implement comprehensive logging for debugging and monitoring purposes, with all log data retained for a minimum of 30 days to support issue investigation and performance analysis.

6.4. Future Enhancements

The following features are considered for future releases and are explicitly not part of the current project scope: social media login via OAuth, multi-language support, AI-based gift recommendations, subscription-based gifting services, and native mobile applications for iOS and Android. These enhancements are deferred to later phases to focus on delivering the core functionality within the current release timeline.